Up to this point we have considered adding waves of the same frequency with different amplitudes and phases. Let's start today by looking at adding waves with the same amplitude and phase but which differ in frequency

$\omega_1 > \omega_2$ and $k_1 = \frac{\omega_1}{v} > k_2 = \frac{\omega_2}{v}$

$$\left.\begin{aligned} \tilde{y}_1(x,t) &= A e^{j(\omega_1 t - k_1 x)} \\ \tilde{y}_2(x,t) &= A e^{j(\omega_2 t - k_2 x)} \end{aligned}\right\} \text{ taking } \phi_1 = \phi_2 = 0 \text{ for convenience.}$$

There is a trick we can employ to make these two waves look similar:

$$\omega_1 = \frac{1}{2}(\omega_1 + \omega_2) + \frac{1}{2}(\omega_1 - \omega_2)$$
$$\omega_2 = \frac{1}{2}(\omega_1 + \omega_2) - \frac{1}{2}(\omega_1 - \omega_2)$$

and similarly for k_1 and k_2

Now we can define $\omega_{av} = \frac{1}{2}(\omega_1 + \omega_2)$ and $\Delta\omega = \omega_1 - \omega_2$

$$\Rightarrow \omega_1 = \omega_{av} + \frac{1}{2}\Delta\omega \quad , \quad \omega_2 = \omega_{av} - \frac{1}{2}\Delta\omega$$
$$k_1 = k_{av} + \frac{1}{2}\Delta k \quad , \quad k_2 = k_{av} - \frac{1}{2}\Delta k$$

We now add these two waves together to yield the resultant wave

$$\tilde{y}(x,t) = A e^{j\left(\omega_{av}t + \frac{1}{2}\Delta\omega t - k_{av}x - \frac{1}{2}\Delta k x\right)} + A e^{j\left(\omega_{av}t - \frac{1}{2}\Delta\omega t - k_{av}x + \frac{1}{2}\Delta k x\right)}$$

$$= A\left(\underbrace{e^{j\left(\frac{1}{2}\Delta\omega t - \frac{1}{2}\Delta k x\right)}}_{\cos(\cdots)+j\sin(\cdots)} + \underbrace{e^{-j\left(\frac{1}{2}\Delta\omega t - \frac{1}{2}\Delta k x\right)}}_{\cos(\cdots)-j\sin(\cdots)}\right) e^{j(\omega_{av}t - k_{av}x)}$$

The terms in brackets can be combined using Euler's formula

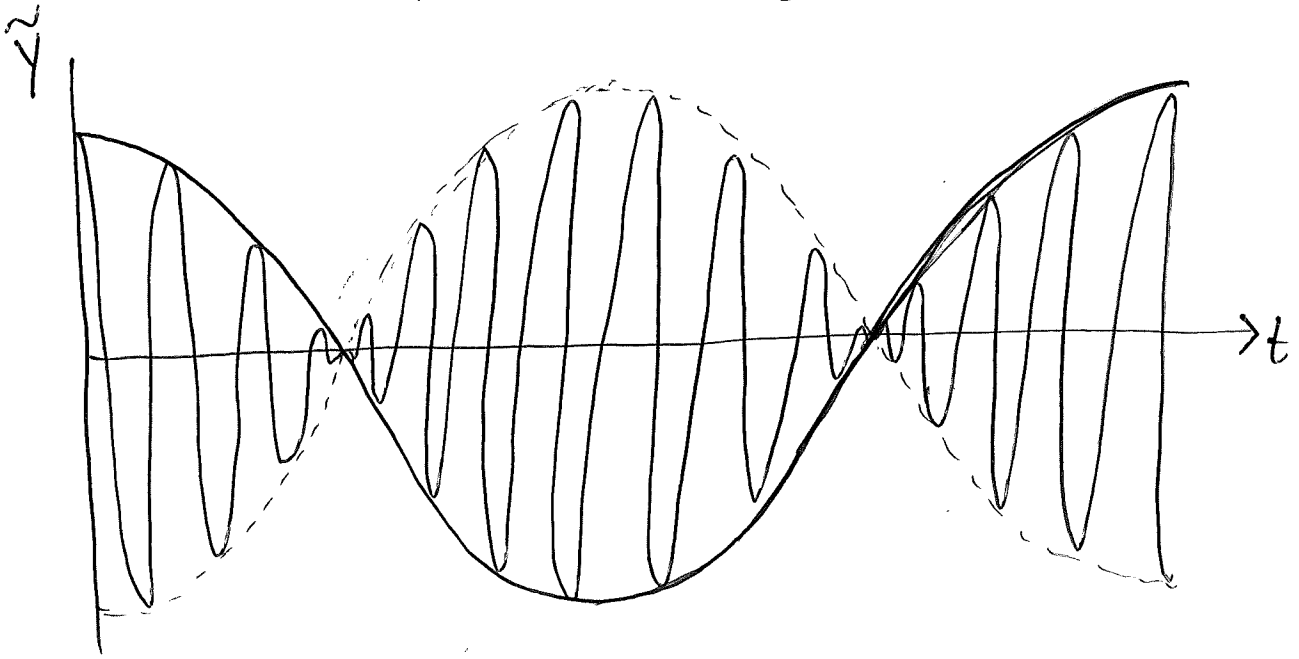
$$\tilde{y}(x,t) = \left[2A\cos\left(\tfrac{1}{2}\Delta\omega t - \tfrac{1}{2}\Delta k x\right)\right] e^{j(\omega_{av}t - k_{av}x)}$$

↑ amplitude is also a wave relying on $\Delta\omega$ and Δk

↑ wave with the average frequency and wavenumber

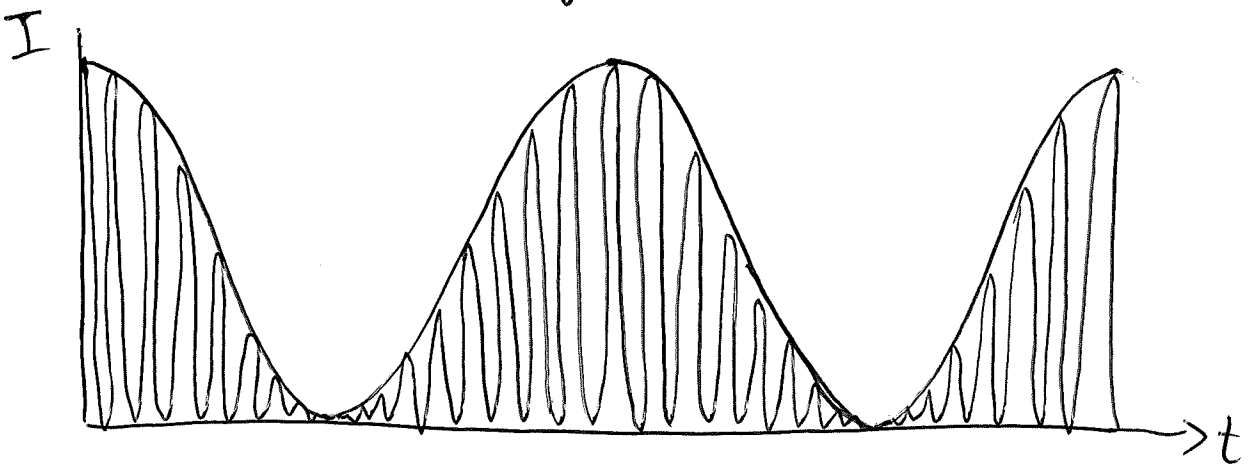
Of particular interest here is when ω_1 and ω_2 are very close together so that $\omega_{av} \gg \Delta\omega$. If we look at a fixed position at $x=0$ for simplicity, the amplitude is oscillating:

$$\tilde{y}(0,t) = \left[2A\cos\left(\tfrac{1}{2}\Delta\omega t\right)\right] e^{j(\omega_{av}t - k_{av}x)}$$



The amplitude wave modulates the amplitude of the average wave by a sinusoidal envelope. When $\cos(\frac{1}{2}\Delta\omega t)$ has a max/min the wave is maximized, and when it has a zero the wave is zero.

Recall the intensity of the wave is the wave squared, so the sound intensity looks like:



Important point: the frequency of $\cos(\frac{1}{2}\Delta\omega t)$ is $\frac{1}{2}\Delta\omega$ but the square of this has <u>twice</u> the frequency, i.e. $\Delta\omega$.

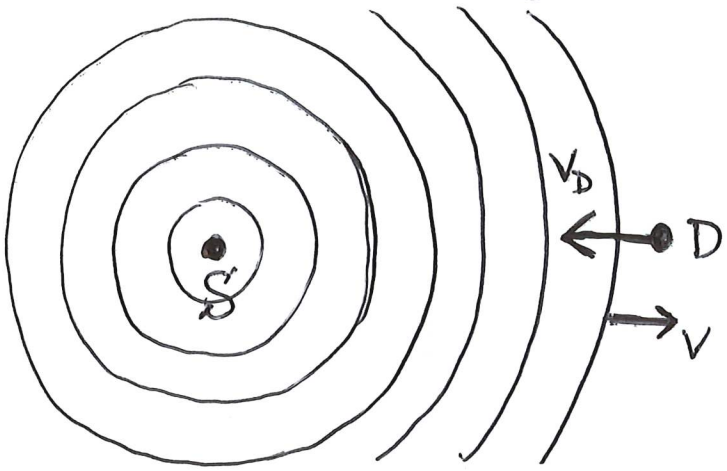
We thus call the <u>beat frequency</u>

$$\omega_{beat} = \Delta\omega \quad \text{or} \quad f_{beat} = \Delta f$$

$\Rightarrow$ Top Hat Question, followed by demo video

5 minute break

Now let's look at how the frequency of a wave can become shifted compared to the source generating the waves, which then leads to numerous applications with the beat frequency. Consider a point source emitting sound in the rest frame of the fluid (i.e. no wind or currents) and a detector moving toward the source at speed v_D



The waves are moving at speed v outward from the source.

The speed of the sound waves from the perspective of the detector is $v' = v + v_D$ but the wavelength is still λ

$$\Rightarrow v' = \lambda f' \quad \text{where} \quad \lambda = \frac{v}{f}$$

$$v + v_D = \frac{v}{f} f' \quad \text{so} \quad \boxed{f' = \left(\frac{v + v_D}{v}\right) f}$$

frequency increases if detector moves toward the source.

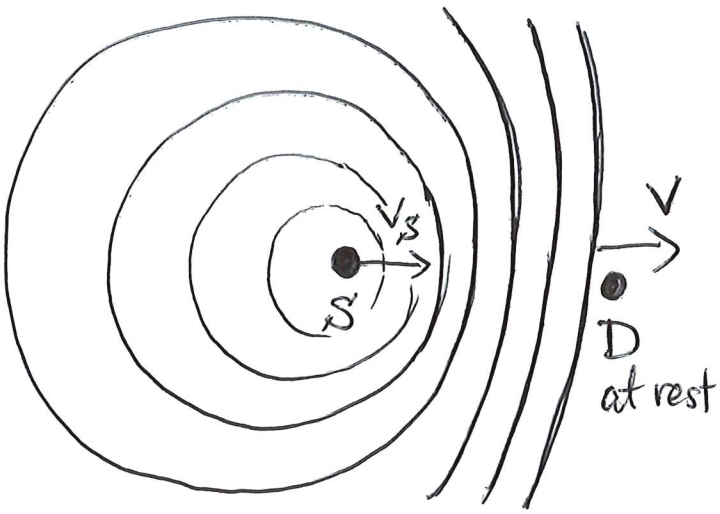
Source stationary, moving detector:

$$\boxed{f' = \left(\frac{v \pm v_D}{v}\right) f}$$

+ : detector moves toward source

− : detector moves away from source.

If the source moves, it is no longer at rest w.r.t. the fluid medium. The waves travel at speed v relative to the medium



The wavelength in direction of source motion is decreased and in opposite direction it is increased

In a time $T = \frac{1}{f}$ the source has moved a distance TV_s

So the wavelength is shifted by $\lambda' = \lambda - TV_s = \lambda - \frac{V_s}{f}$

but $\lambda = \frac{V}{f}$ and $\lambda' = \frac{V}{f'}$

$$\Rightarrow \frac{V}{f'} = \frac{V}{f} - \frac{V_s}{f} \quad \text{so} \quad \boxed{f' = \left(\frac{V}{V - V_s}\right) f}$$

frequency increases when source moves toward detector

Moving source, stationary detector:

$$f' = \left(\frac{V}{V \pm V_s}\right) f$$

+: source moving away from detector
−: source moving toward detector

Combining the two together for the most general case gives

$$\boxed{f' = \left(\frac{V \pm V_D}{V \pm V_s}\right) f}$$

$\Rightarrow$ Top Hat Q

Example 1: How fast would you need to move toward a stationary sound source to make its apparent frequency increase by 50%? (Use $v = 344$ m/s)

$$f' = \left(\frac{v \pm v_D}{v \pm v_S}\right) f$$

here $v_S = 0$ and D is moving toward S, so numerator has + sign

$$f' = \left(\frac{v + v_D}{v}\right) f \Rightarrow v_D = \frac{f'}{f} v - v = \left(\frac{f' - f}{f}\right) v$$

Therefore $v_D = \left(\frac{1.5f - f}{f}\right)(344\text{ m/s}) = (0.5)(344\text{ m/s}) = \boxed{172\text{ m/s}}$

Example 2: How fast would a sound source need to move toward you to make its apparent frequency increase by 50%?

Here $v_D = 0$ and S moving toward D so denominator has − sign:

$$f' = \frac{v}{v - v_S} f \Rightarrow v - v_S = v\frac{f}{f'} \quad \text{so} \quad v_S = v\left(\frac{f' - f}{f'}\right)$$

Therefore $v_S = \left(\frac{1.5f - f}{1.5f}\right) v = \left(\frac{1/2}{3/2}\right) v = \frac{1}{3} v = \boxed{115\text{ m/s}}$

Take home: the two are <u>not</u> symmetric. Think about why that is true.